DNA – find out for yourself

The ability to find relevant information, critically analyse it and summarise what it tells you are important to every scientist. Now it's time to test your skills at this process. You need access to the Internet and/or biology and genetics resources to complete this task.

1 Each box below and on the next page contains key words relating to an important discovery about DNA. Use the resources supplied or the Internet to:

- describe what was discovered or achieved
- explain how this contributed to human understanding of DNA.

Record the information described above in note form in each box, and remember to record details of the source of your information (i.e. book title and page number, or website).

1865, Gregor Mendel, dominant, recessive, characteristics

1902, Walter Sutton, chromosomes, cell division

1913, Alfred Sturtevant, genetic map, chromosomes

1950, Barbara McClintock, genes, characteristics

1952, Alfred Hershey, Martha Chase, experiment, genes

1953, James Watson, Francis Crick, Maurice Wilkins, Rosalind Franklin, chemical structure

1956, Jo Hin Tjio, Albert Levan, 46, human

1958, Matthew Meselson, Franklin Stahl, DNA, replicated

1967, Har Gobind Khorana, Marshall Nirenberg, amino acids

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1970, restriction enzymes, DNA

1977, nucleotides, DNA molecule, virus, Fred Sanger, Walter Gilbert, Allan Maxam

1978, Louise Brown, IVF

1978, Genentech, gene, insulin

1984, Alec Jeffries, DNA sequence

1986, GM crops, herbicide-resistant

1989, human gene, Francis Collins, Lap-Chee Tsui, cystic fibrosis.

1997, Hunt Willard, artificial, chromosome

1997, Roslyn Institute, Dolly, sheep

1998, Dolly, Bonnie

2003, The Human Genome Project

2 Now use a word-processing program or a large sheet of poster paper to create a timeline including the events listed above. Also include any other events you think are significant to the development of our understanding of DNA. Where possible, add labelled diagrams or illustrations relevant to each event.

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